

5.4.3 Planetary motion

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) Kepler's three laws of planetary motion	HSW7
(b) the centripetal force on a planet is provided by the gravitational force between it and the Sun	
(c) the equation $T^2 = \left(\frac{4\pi^2}{GM}\right)r^3$	Learners will also be expected to derive this equation from first principles. HSW1
(d) the relationship for Kepler's third law $T^2 \propto r^3$ applied to systems other than our solar system	
(e) geostationary orbit; uses of geostationary satellites.	HSW1, 2, 9, 10 Predicting geostationary orbit using Newtonian laws.